

THIRUMALAI G

AI / ML ENGINEER

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI/ML engineer with 1.5+ years of hands-on experience building AI agents, RAG systems, and web applications powered by LLMs. Skilled in developing and fine-tuning models, designing automation workflows, and delivering projects independently or as part of a collaborative team.

TECHNICAL SKILLS

AI & ML: Large Language Models (LLM), Vision Language Models (VLM), AI Agents, AI-Assisted Development, AI Automation

Programming: Python, Command Line Interface (CLI)

Vector DBs & Tools: LangChain, PyTorch, ChromaDB, Hugging Face Transformers, OpenCV, NumPy, Pandas, Git, Linux, Hugging Face Spaces

Workflow & API: n8n, ComfyUI, OmniRoute, Claude Code

WORK EXPERIENCE

AI Developer | ZFusionAI Tech Service | Tamil Nadu, India

Jun 2025 – May 2026

- **AI Agent & RAG System:** Developed AI agents to complete small tasks without relying on paid APIs, using a RAG architecture and vector stores.
- **AI Automation:** Built a full frontend web-app builder combining semi-agentic and rule-based automation. [Source code](#).
- **AI Video Generation Pipeline (Free Resources):** Built a video generation pipeline using only free resources, enabling AI-generated story videos, ads, and similar content. [Source code](#).
- **ComfyUI Custom Nodes:** Developed multiple reusable ComfyUI nodes for character consistency, enabling stories to be generated with the same character while sourcing copyright-safe reference imagery. Available on GitHub.
- **Cloud MLOps:** Built an end-to-end MLOps pipeline covering automated dataset ingestion, feature engineering, model training, performance visualization, and model export.

AI Intern | Vcodez Tech Service | Tamil Nadu, India

May 2025 – Jun 2025

- **Machine Learning & Generative AI:** Contributed to multiple ML and generative AI projects, including a real-estate visualization solution built with Stable Diffusion (SD) and LoRA (Low-Rank Adaptation) fine-tuning.
- **AI Sign Language Classification:** Developed a sign language classification model using a Graph Neural Network (GNN) to support seamless communication with deaf individuals. [Source code](#).

SIGNATURE PROJECT

Kratos Agent — Autonomous Terminal AI Coding Agent

[Source code](#)

- **Overview:** Built a full autonomous terminal-based AI coding agent in Python that plans, writes, edits, and verifies code end-to-end with minimal human intervention.
- **Universal LLM Brain:** Designed a proxy-free gateway layer with SSE streaming and a multi-format tool-call parser (Harmony/OpenAI-OSS, Hermes/Qwen/Claude XML, JSON) for compatibility across LLM backends.
- **Autonomous Tool Forging:** Built a self-extending tool system where the agent writes new Python tools on demand, registers their schemas, and hot-reloads them into the live session.
- **Planning & Memory:** Implemented a dynamic step planner with live task tracking and isolated multi-session memory supporting instant switching and history persistence.
- **Engineering Workflow Suite:** Integrated senior-level review, simplification, and security-audit commands, 695+ indexed Claude Code prompts/sub-agents, 148+ skills, and 48+ passing tests.
- **Execution Safety:** Implemented tiered approval modes (suggest, auto-edit, full-auto) to control the agent's autonomy over file edits and terminal commands.

EDUCATION

Bachelor of Technology (B.Tech), Artificial Intelligence & Data Science

Muthayammal College of Engineering | 2021 – 2025